

Monte-Carlo Prediction for Action Values?

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- 1: Initialize $v_\pi(s), \forall s \in S \longrightarrow q_\pi(s,a), \forall s \in S, \forall a \in A$.
 - 2: **while** true **do**
 - 3: Randomly pick an initial state S_0
 - 4: Generate an episode using π . We get
 - 5: $S_0, \underline{A_0}, \underline{R_1}, S_1, A_1, R_2, \dots, S_T, R_T$
 - 6: $G = 0$
 - 7: **for** $t = T - 1, T - 2, \dots, 0$ **do**
 - 8: $G = G + \gamma R_{t+1}$
 - 9: $v_\pi(S_t) = (1 - \alpha)v_\pi(S_t) + \alpha G$
-

$$q_\pi(S_t, A_t) = (1 - \alpha)q_\pi(S_t, A_t) + \alpha G.$$

Recall Going From State to Action Value

$$q_{\pi}(s, a) = E_{\pi} \left[R_{t+1} + \gamma V_{\pi}(S_{t+1}) \mid S_t = s, A_t = a \right]$$

$$V_{\pi}(s) = \sum_a q_{\pi}(s, a) \pi(a|s).$$

- What if the model isn't known?
- How about action to state value?

Exploration

- The policy may not visit certain state-action pairs during the episodes generated using it
- Slows down evaluation
- Force exploration using *exploring starts*

Monte-Carlo Control

- We want to find the optimal action value function and the optimal policy
- MC with Policy Iteration

Monte-Carlo Control

- We want to find the optimal action value function and the optimal policy
- We will use GPI... We expect
 - Policy Evaluation (abridged)
 - Policy Improvement

Monte-Carlo Control

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1:  $\pi = \pi_0$ 
2: while true do
3: Randomly pick an initial state  $S_0$  and action  $A_0$ 
4: Generate an episode using  $\pi$ . We get
5:    $S_0, A_0, R_1, S_1, A_1, R_2, \dots, S_T, R_T$ 
6:   for  $t = T - 1, T - 2, \dots, 0$  do
7:      $G = G + \gamma R_{t+1}$ 
8:      $Q_\pi(S_t, A_t) = (1 - \alpha)Q_\pi(S_t, A_t) + \alpha G$ 
9:      $\pi(S_t) = \underset{a \in \mathcal{A}(S_t)}{\operatorname{argmax}} Q_\pi(S_t, a)$ 

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Monte-Carlo Control

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1:  $\pi = \pi_0$ 
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8:      $Q_\pi(S_t, A_t) = (1 - \alpha)Q_\pi(S_t, A_t) + \alpha G$ 
9:      $\pi(S_t) = \arg \max_{a \in A(S_t)} Q_\pi(S_t, a)$ 

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Exploration

- State and action values visited must not be entirely biased by the current policy
- The policy may not visit certain state-action pairs during the episodes generated using it
 - Greedy policy and the target for the next evaluation in case of control
 - The target policy being evaluated in case of prediction
- Hampers exploration and slows down evaluation
- Force exploration using *exploring starts*

Exploration

- Exploring starts
 - May not be possible for agent to do in the real environment
- In the absence of exploring starts
 - On-Policy exploration
 - Agent uses policy being evaluated (**target**) during the episode
 - Off-Policy exploration
 - Agent may use a different **behavior** policy to generate the episode than the **target** policy

On-Policy

- Restrict the agent to a set of policies that have exploration inbuilt
- Consider a set of policies called ϵ -soft policies

$$\pi(a|s) \geq \frac{\epsilon}{|A(s)|}, \forall a \in A(s)$$

~~$$\pi(a|s) \leq 1 - \epsilon, a = \arg \max_j Q(s, j)$$~~

On-Policy

- An example is the ϵ -greedy policy:

$$\pi(a|s) = \begin{cases} 1 - \epsilon & a = \arg \max_j Q(s, j) \\ \frac{\epsilon}{|A(s)|} & \forall a \in A(s), a \neq \arg \max_j Q(s, j) \end{cases}$$

- With what probability do we choose the greedy action?

How Does This Change MC Control?

- We have Policy Evaluation and Improvement as per GPI
- An ϵ -greedy policy obtained from evaluation of an ϵ -soft is at least as good!
- If there is no improvement, then you have the optimal ϵ -soft policy

Book ↓ has the proof.

Monte-Carlo Control: Make it On-Policy?

1: $\pi = \pi_0$

2: **while** true **do**

3: Randomly pick an initial state S_0 and action A_0

4: Generate an episode using π . We get

5: $S_0, A_0, R_1, S_1, A_1, R_2, \dots, S_T, R_T$

6: **for** $t = T - 1, T - 2, \dots, 0$ **do**

7: $G = G + \gamma R_{t+1}$

8: $Q_\pi(S_t, A_t) = (1 - \alpha)Q_\pi(S_t, A_t) + \alpha G$

9: $\pi(S_t) = \arg \max_{a \in A(S_t)} Q_\pi(S_t, a)$

Handwritten notes:

- $\pi()$ is both the behavior & the target.
- replace by $\pi(S_t)$ that is ϵ -greedy.
- $= \begin{cases} 1 - \epsilon + \frac{\epsilon}{|A(s)|} a = \arg \max & \dots \\ \frac{\epsilon}{|A(s)|} & \text{other actions.} \end{cases}$

How Do We Show This?

- An ϵ -greedy policy obtained from evaluation of an ϵ -soft is at least as good!
- If there is no improvement, then you have the optimal ϵ -soft policy

Off-Policy

- Use behavior policy with good exploration properties to generate episodes
- Must use episodes to estimate action-value function of **target** policy
- On-policy is a subset of off-policy

Off-Policy

- Use behavior policy with good exploration properties to generate episodes
- Must use episodes to estimate action-value function of **target** policy
- On-policy is a subset of off-policy
- Off-policy may be slower to converge
- Allows learning from data generated by an expert

Off-Policy

- Further require to satisfy the assumption of coverage

$$b(a|s) \neq 0, \text{ for } a \in \{a : \pi(a|s) \neq 0\}$$

Off-Policy

- We are interested in expected returns
- Since episodes are generated using behavior policy:
 - Express expected return using target in terms of expected return when using behavior

$$E_{\pi}[G_t|S_t] = E_b[\rho_{t:T-1}G_t|S_t]$$

- Importance sampling factor (weight obtained return by this factor)

$$\rho_{t:T-1} = \prod_{k=t}^{T-1} \frac{\pi(a_k|s_k)}{b(a_k|s_k)}$$

$$E_{\pi}[C_t | S_t = s_t] = V_{\pi}(s_t)$$

$$= \sum_{(s_{t+1}, \dots, s_{t+T})} (r_{t+1} + \gamma r_{t+2} + \dots + \gamma^{T-t} r_{t+T}) P_{\pi}[R_{t+1}=r_{t+1}, R_{t+2}=r_{t+2}, \dots, R_{t+T}=r_{t+T} | S_t=s_t]$$

$$= \sum_{(s_{t+1}, \dots, s_{t+T})} \sum_{(a_{t+1}, \dots, a_{t+T})} (r_{t+1} + \dots + r_{t+T}) P_{\pi}[S_{t+1}=s_{t+1}, R_{t+1}=r_{t+1}, \dots, S_{t+T}=s_{t+T}, R_{t+T}=r_{t+T} | S_t=s_t]$$

$$= \sum_{a_t \in A(s)} \sum_{(s_{t+1}, \dots, s_{t+T})} \sum_{(a_{t+1}, \dots, a_{t+T})} (r_{t+1} + \dots + r_{t+T}) P_{\pi}[S_{t+1}=s_{t+1}, R_{t+1}=r_{t+1}, \dots, S_{t+T}=s_{t+T}, R_{t+T}=r_{t+T} | S_t=s_t, A_t=a_t] \pi(a_t | s)$$

$$= \sum_{(a_t, a_{t+1}, \dots, a_{t+T-1})} \sum_{\bar{s}} \sum_{\bar{r}} P_{\pi}[S_{t+1}=s_{t+1}, R_{t+1}=r_{t+1}, A_{t+1}=a_{t+1}, \dots, S_{t+T-1}=s_{t+T-1}, A_{t+T-1}=a_{t+T-1}, R_{t+T-1}=r_{t+T-1}, S_{t+T}=s_{t+T}, R_{t+T}=r_{t+T}, A_{t+T}=a_{t+T} | S_t=s_t, A_t=a_t] \pi(a | s)$$

$$P_{\pi}[S_{t+1}=s_{t+1}, R_{t+1}=r_{t+1}, A_{t+1}=a_{t+1}, \dots, S_{t+T-1}=s_{t+T-1}, A_{t+T-1}=a_{t+T-1}, R_{t+T-1}=r_{t+T-1}, S_{t+T}=s_{t+T}, R_{t+T}=r_{t+T}, A_{t+T}=a_{t+T} | S_t=s_t, A_t=a_t] \pi(a | s)$$

$$= P[S_{t+1}=s_{t+1}, R_{t+1}=r_{t+1} | S_t=s_t, A_t=a_t, S_{t+2}=s_{t+2}, A_{t+2}=a_{t+2}, R_{t+2}=r_{t+2}, \dots, S_{t+T-1}=s_{t+T-1}, A_{t+T-1}=a_{t+T-1}, R_{t+T-1}=r_{t+T-1}, S_{t+T}=s_{t+T}, A_{t+T}=a_{t+T}, R_{t+T}=r_{t+T}] \pi(a_t | s_t)$$

$$= P[S_{t+1}=s_{t+1}, R_{t+1}=r_{t+1} | S_t=s_t, A_t=a_t, S_{t+2}=s_{t+2}, A_{t+2}=a_{t+2}, R_{t+2}=r_{t+2}, \dots, S_{t+T-1}=s_{t+T-1}, A_{t+T-1}=a_{t+T-1}, R_{t+T-1}=r_{t+T-1}] \leftarrow \text{Using the Markov property}$$

$$\times P[A_{t+T-1}=a_{t+T-1} | S_{t+T-1}=s_{t+T-1}, R_{t+T-1}=r_{t+T-1}, \dots, R_{t+1}=r_{t+1}, S_t=s_t, A_t=a_t]$$

$$\times P[S_{t+T-1}=s_{t+T-1}, R_{t+T-1}=r_{t+T-1}, \dots, R_{t+1}=r_{t+1} | S_t=s_t, A_t=a_t] \pi(a_t | s_t)$$

$$= p(s_{t+1}, r_{t+1} | s_t, a_t) \pi(a_{t+1} | s_{t+1}) \dots p(s_{t+T-1}, r_{t+T-1} | s_{t+T-2}, a_{t+T-2}) \pi(a_{t+T-1} | s_{t+T-1}) \dots p(s_{t+1}, r_{t+1} | s_t, a_t) \pi(a_t | s_t)$$

$$E_{\pi}[G_t | S_t = s_t] = V_{\pi}(s_t)$$

$$= \sum_{(a_{t+1}, \dots, a_{t+T})} (r_{t+1} + \gamma r_{t+2} + \dots + \gamma^{T-1} r_{t+T}) P_{\pi}[R_{t+1}=r_{t+1}, R_{t+2}=r_{t+2}, \dots, R_{t+T}=r_{t+T} | S_t = s_t]$$

$$= \sum_{(a_{t+1}, \dots, a_{t+T})} \sum_{(s_{t+1}, \dots, s_{t+T})} (r_{t+1} + \dots) P_{\pi}[S_{t+1}=s_{t+1}, R_{t+1}=r_{t+1}, \dots, S_{t+T}=s_{t+T}, R_{t+T}=r_{t+T} | S_t = s_t]$$

$$= \sum_{a_t \in A(s_t)} \sum_{(a_{t+1}, \dots, a_{t+T})} \sum_{(s_{t+1}, \dots, s_{t+T})} (r_{t+1} + \dots) P_{\pi}[S_{t+1}=s_{t+1}, R_{t+1}=r_{t+1}, \dots, S_{t+T}=s_{t+T}, R_{t+T}=r_{t+T} | S_t = s_t, A_t = a_t] \pi(a_t | s_t)$$

$$= \sum_{(a_t, a_{t+1}, \dots, a_{t+T-1})} \sum_{s_t} \sum_{s_t} (r_{t+1} + \dots) P_{\pi}[S_{t+1}=s_{t+1}, R_{t+1}=r_{t+1}, A_{t+1}=a_{t+1}, \dots, S_{t+T-1}=s_{t+T-1}, A_{t+T-1}=a_{t+T-1}, R_{t+T-1}=r_{t+T-1}, S_{t+T}=s_{t+T}, R_{t+T}=r_{t+T} | S_t = s_t, A_t = a_t] \pi(a_t | s_t)$$

$$= \sum_{(a_t, a_{t+1}, \dots, a_{t+T-1})} \sum_{s_t} \sum_{s_t} (r_{t+1} + \dots) p(s_{t+1}, a_{t+1} | s_t, a_t) \pi(a_{t+1} | s_{t+1}) \dots p(s_{t+T-1}, a_{t+T-1} | s_{t+T-2}, a_{t+T-2}) \pi(a_{t+T-1} | s_{t+T-1})$$

$$\underbrace{\dots p(s_{t+1}, a_{t+1} | s_t, a_t) \pi(a_{t+1} | s_{t+1})}_{\prod_{k=0}^{T-1} p(s_{t+k}, a_{t+k} | s_{t+k-1}, a_{t+k-1})}$$

$$E_{\pi}[G_t | S_t = s_t] =$$

$$\sum_{a_t} \dots \sum_{s_t} (r_{t+1} + \dots) \left(\frac{\prod_{k=0}^{T-1} \pi(a_{t+k} | s_{t+k})}{\prod_{k=0}^{T-1} b(a_{t+k} | s_{t+k})} \right) \prod_{k=1}^{T-1} p(s_{t+k}, a_{t+k} | s_{t+k-1}, a_{t+k-1}) \frac{\prod_{k=0}^{T-1} b(a_{t+k} | s_{t+k})}{\prod_{k=0}^{T-1} b(a_{t+k} | s_{t+k})}$$

$$= E_b \left[\overset{\text{IS factor}}{\downarrow} G_t | S_t = s \right]$$

Off-Policy

- Importance sampling factor (weight obtained return by)

$$\rho_{t:T-1} = \prod_{k=t}^{T-1} \frac{\pi(a_k | s_k)}{b(a_k | s_k)}$$

- Scale the G obtained by the factor in the policy evaluation step

Off-Policy

- Importance sampling factor (weight obtained return by)

$$\rho_{t:T-1} = \prod_{k=t}^{T-1} \frac{\pi(a_k | s_k)}{b(a_k | s_k)}$$

- Scale the G obtained by the factor in the policy evaluation step
- Average Return?
 - Suppose we obtain $K(s)$ returns for state s

$$v_{\pi}(s) \approx \frac{1}{K(s)} \sum_{k=1}^{K(s)} \rho_{t(k):T(k)-1} G^{(k)}$$

Off-Policy: Calculating Average Return

- Suppose we obtain $K(s)$ returns for state s
- Ordinary importance sampling
- Weighted importance sampling

$$\mathbb{E}_b \left[\frac{1}{K(s)} \sum_{k=1}^{K(s)} \rho_{t(k):T(k)-1} G^{(k)} \right]$$

An unbiased estimate of $v_\pi(s)$, since its expectation is equal for $v_\pi(s)$.

$$v_\pi(s) \approx \frac{\sum_{k=1}^{K(s)} \rho_{t(k):T(k)-1} G^{(k)}}{\sum_{k=1}^{K(s)} \rho_{t(k):T(k)-1}}$$

$$\begin{aligned} &= \frac{1}{K(s)} \sum_{k=1}^{K(s)} \mathbb{E}_b \left[\rho_{t(k):T(k)-1} G^{(k)} \right] \\ &= \frac{1}{K(s)} \sum_{k=1}^{K(s)} \mathbb{E}_\pi \left[G^{(k)} \right] \\ &= v_\pi(s). \end{aligned}$$

Ordinary IS vs. Weighted IS

- Assume first-visit MC
- Ordinary importance sampling

$$v_{\pi}(s) \approx \frac{1}{K(s)} \sum_{k=1}^{K(s)} \rho_{t(k):T(k)-1} G^{(k)}$$

Can be very large. Resulting
a very large

- Weighted importance sampling

$$v_{\pi}(s) \approx \frac{\sum_{k=1}^{K(s)} \rho_{t(k):T(k)-1} G^{(k)}}{\sum_{k=1}^{K(s)} \rho_{t(k):T(k)-1}}$$

$\hookrightarrow E(0,1)$

Variance
of $PG^{(n)}$

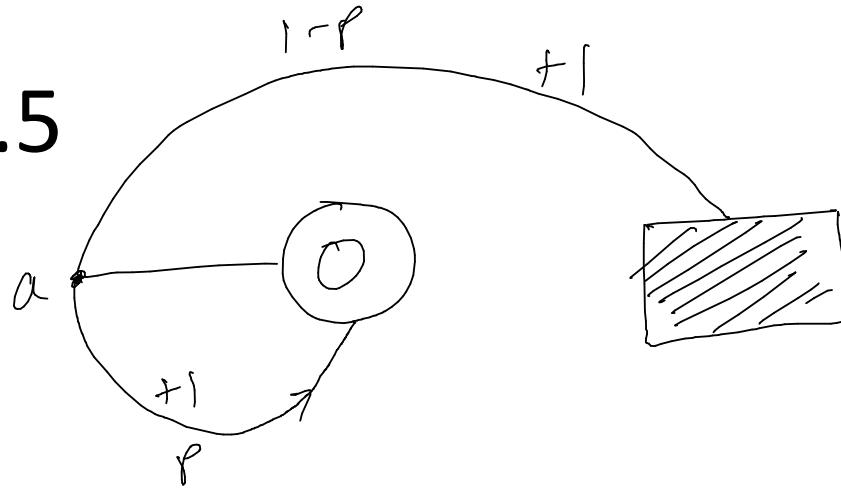
First-visit vs Every-visit

- Ordinary importance sampling

$$v_{\pi}(s) \approx \frac{1}{K(s)} \sum_{k=1}^{K(s)} \rho_{t(k):T(k)-1} G^{(k)}$$

Exercise 5.5

(1) $V_{\pi}(0) = ?$



$$V_{\pi}(0) =$$

$$E[G_t | S_t = 0]$$

(2) Given episode of length n , calculate first-visit & every visit MC estimates of value.

6 1 1 1 - - - 1 (n times).

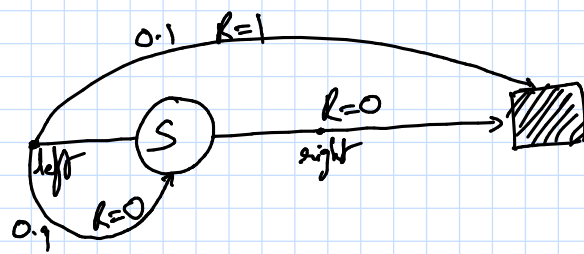
$$\hat{V}_{\pi}(0) = n \quad (\text{first-visit})$$

$$\hat{V}_{\pi}(0) = \frac{n+1}{2} \quad (\text{every-visit})$$

$$G_t = \begin{cases} 1 & 1-p \\ 2 & p(1-p) \\ \vdots & \end{cases}$$

$$V_{\pi}(0) = \frac{1}{1-p}$$

Expected Value and Variance [Ordinary & Weighted IS]



$$\pi(\text{left}|s) = 1 \quad [\text{Target}]$$

$$b(\text{left}|s) = 1/2 \quad [\text{Behavior}]$$

$$E[G_T | S_T = s] = E_b[P_{t:T-1} G_T | S_T = s]. \quad \left[\begin{array}{l} \text{We showed} \\ \text{this} \end{array} \right]$$

In practice, for state s , we will have one or more sample returns G_t obtained using behavior policy $b(\cdot|\cdot)$.

We will use $P_{t:T-1} G_T$ to update our estimate of $V_\pi(s)$.

$$\hat{V}_\pi(s) = (1-\alpha) \hat{V}_\pi(s) + \alpha P_{t:T-1} G_T.$$

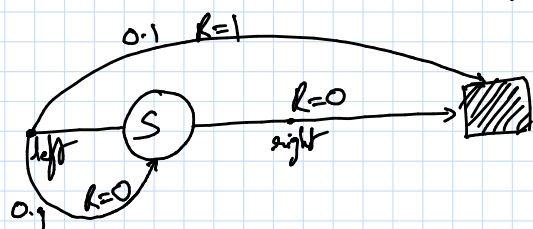
→ For when $\hat{V}_\pi(s)$ is the sample mean, the above is an unbiased estimate of $V_\pi(s)$. That is $E[\hat{V}_\pi(s)] = V_\pi(s)$.

→ However, variance of the estimate matters.

↳ Variance of $P_{t:T-1} G_T$ is key.

↳ Captures how "noisy" the estimate is.

Expected Value and Variance [Ordinary & Weighted IS]



$$\pi(\text{left}|s) = 1 \quad [\text{Target}]$$

$$b(\text{left}|s) = \frac{1}{2} \quad [\text{Behavior}]$$

$$E_b[p_{0:T-1} G_0] = ?$$

$$T=1: \left\{ E_b[p_{0:0} G_0] = 1 \quad p_{0:0} = 1 \quad b(\text{left}|s) (0.1) = \frac{\pi(\text{left}|s)}{b(\text{left}|s)} b(\text{left}|s) (0.1) \right.$$

$$T=2: \left\{ E_b[p_{0:1} G_0] = 1 \quad p_{0:1} = 1 \quad \frac{\pi(\text{left}|s) \pi(\text{left}|s)}{b(\text{left}|s) b(\text{left}|s)} b(\text{left}|s) b(\text{left}|s) (0.9)(0.1) \right.$$

$$\sum \downarrow = 0.1 + 1(0.9)(0.1) + 1(0.9)^2(0.1) + \dots$$

$$= 0.1 \left(\frac{1}{1-0.9} \right) = 1.$$

$$E_b[(p_{0:T-1} G_0)^2] = \dots$$

Reading

Sections 5.6 and 5.7 on Off-Policy MC prediction and control respectively

Temporal Difference Learning

Temporal Difference Learning

- Bootstrap using prior knowledge like DP

$$\begin{aligned} v_{\pi_k}(s) &= E[R_{t+1} + \gamma v_{\pi_k}(S_{t+1}) | S_t = s] \\ &= \sum_a \pi(a|s) \sum_{s', r} (r + \gamma v_{\pi_k}(s')) p(s', r | s, a) \end{aligned}$$

- However, use sample estimates of expectation like MC
 - Learn from experience
- Method, unlike MC, is truly online

Averaging in MC

You ran an episode, obtained a return and used it to update the value function (expected return starting in the state S_t)

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4: Generate an episode using  $\pi$ . We get
5:    $S_0, A_0, R_1, S_1, A_1, R_2, \dots, S_T, R_T$ 
6:    $G = 0$ 
7:   for  $t = T - 1, T - 2, \dots, 0$  do
8:      $G = G + \gamma R_{t+1}$ 
9:      $v_\pi(S_t) = (1 - \alpha)v_\pi(S_t) + \alpha G$ 

```


Temporal Difference.... You are Impatient

- Perform action at time t ,
- To update value of state at t
 - Use the resulting reward,
 - and Bootstrap using current value of the next state!

$$\begin{aligned}
 v_{\pi}(S_t) &= (1 - \alpha)v_{\pi}(S_t) + \alpha(R_{t+1} + \gamma v_{\pi}(S_{t+1})) \\
 &= v_{\pi}(S_t) + \alpha(R_{t+1} + \gamma v_{\pi}(S_{t+1}) - v_{\pi}(S_t))
 \end{aligned}$$